THE BASE CHANGE OF THE SHEAF WITH EXACT SUPPORT IS AN ISOMORPHISM EXACTLY WHEN THE CORRESPONDING BASE CHANGE MAP IS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2L (sources/stacks-duality/pinned-original-source.tex line 2561).

Allowed standard prerequisites (editorial).

- Base change for the sheaf with exact support and the fancy base change map. Same chapter, equation labelled equation-base-change-exact-support, invoked in the statement, and Cohomology, Remark labelled cohomology-remark-prepare-fancy-base-change, cited in the statement and at proof lines 2610-2612. Fixes the two maps being compared.
- Base change for a closed immersion under tor independence. Derived Categories of Schemes, Lemma labelled perfect-lemma-compare-base-change-closed-immersion, cited at proof lines 2601-2606. Gives Ri'_* o Lg^* = Lf^* o Ri_*.
- The description of the pushed forward sheaf with exact support. Same chapter, Lemma labelled lemma-sheaf-with-exact-support-ext, cited at proof lines 2606-2610. Identifies source and target with the objects of the statement.

Source reference. Original equation 0E2K: sources/stacks-duality/pinned-original-source.tex, source lines 2533–2533. Same-chapter pointer into the author environment equation at source line 2533; invoked at proof line 2577 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 08I3: sources/stacks-chow/tags.txt, source lines 10801–10801. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2591 of the counted proof.

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Source reference. Original lemma 0E23: sources/stacks-chow/tags.txt, source lines 17593–17593. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2581 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A77: sources/stacks-duality/pinned-original-source.tex, source lines 2339–2339. Same-chapter pointer into the author environment lemma at source line 2339; invoked at proof line 2589 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In the situation above, the map (0E2K) is an isomorphism if and only if the base change map*

$$Lf^*R\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_Z, K) \longrightarrow R\mathcal{H}om_{\mathcal{O}_{X'}}(\mathcal{O}_{Z'}, Lf^*K)$$

of Cohomology, Remark 08I3 is an isomorphism.

Proof. The statement makes sense because $\mathcal{O}_{Z'} = Lf^*\mathcal{O}_Z$ by the assumed tor independence. Since i'_* is exact and faithful we see that it suffices to show the map (0E2K) is an isomorphism after applying Ri'_*. Since $Ri'_* \circ Lg^* = Lf^* \circ Ri_*$ by the assumed tor independence and Derived Categories of Schemes, Lemma 0E23 we obtain a map

$$Lf^*Ri_*R\mathcal{H}om(\mathcal{O}_Z, K) \longrightarrow Ri'_*R\mathcal{H}om(\mathcal{O}_{Z'}, Lf^*K)$$

whose source and target are as in the statement of the lemma by Lemma 0A77. We omit the verification that this is the same map as the one constructed in Cohomology, Remark 08I3. □